MIL-M-46312B(MU)
15 August 1972
SUPERSEDING
MIL-M-46312A(MU)
14 February 1964

MILITARY SPECIFICATION

MOUNT, TELESCOPE: M146

1. SCOPE

1.1 This specification covers one type of telescope mount for support of a direct fire telescope (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPCIFICATIONS

Military

| | |
|---|---|
| MIL-F-13926 | Fire Control Materiel: General Specification Governing the Manufacture and Inspection of. |
| MIL-P-14232 | Parts, Equipment and Tools for Ordnance Materiel: Packaging of. |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for inspection by Attributes. |
| MIL-STD-109 | Quality Assurance Terms and Definitions. |

DRAWINGS

U. S. ArMy Frankford Arsenal

| | |
|---|---|
| F8616011 | Mount, Telescope: M146. |

PACKAGING DATA SHEET

| | |
|---|---|
| 8616011 | Packaging of Mount, Telescope: M146. |

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting office.)

FSC 1240

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## 3. REQUIREMENTS

3.1 Fabrication. - The mount shall be manufactured in accordance with Drawing F8616011 and drawings pertaining thereto.

3.2 Temperature conditions.- Except as specified in 3.5.2, the requirements of this specification shall be met at a standard ambient temperature of 73°F ± 18°F.

3.3 General specification.- The following requirements of MIL-F-13926 shall be applicable as dictated by the pertinent design characteristics l

(a) order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Electrical and electronic assemblies
(e) Workmanship

3.4 Orientation.- Whenever specified herin, the following conditions shall apply:

(a) The mount shall be mounted and positioned on the mounting surfaces and keys provided, and immobilized utilizing eight (8) mounting screws and washers.
(b) The mounting surfaces shall be positioned horizontal for performance and operability measurements.
(c) The elevation and azimuth knobs shall be in approximately the center of travel.
(d) A dummy load, weighing 44 lbs. ± 1/4 lb., shall be assembled and secured to the mount with the tapered pin provided. The forward end of the dummy load shall contain a spherical diameter which fits within the three (3) contact points of the cylindrical opening of Arm 8616020; the rear end shall extend beyond the tapered pin an appropriate distance which will permit meeting the requirement of 3.6.2.

3.5 Environmental.

3.5.1 Storage temPeratures.- There shall be no evidence of damage or deterioration of performance due to storage at temperatures of -80°F and +160°F.

3.5.2 Operating temPeratures.- The mount shall meet the applicable requirements of 3.7 while thermally stabilized at ambient temperatures of -40°F and +150°F.

3.5.3 Vibration. - The mount shall show no evidence of physical failure or damage shall meet the requirements of 3.6 to 3.10 inclusive after being vibrated at the conditions specified in 3.5.3.1 or 3.5.3.2.

3.5.3.1 Condition A. - With the mount mounted as specified in 3.4(a) and with the dummy load assembled as specified in 3.4(d), it



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shall be vibrated for 5 minutes ± 15 seconds along each of the three (3) mutually perpendicular axes at .125 inch double amplitude and a constant frequency of 30 cycles per second (cps).

3.5.3.2 Condition B. - With the mount mounted as specified in 3.4(a) and with the dummy load assembled as specified in 3.4(d), it shall be vibrated for 100 minutes ± 1 minute along each of the three (3) mutually perpendicular axes at 0.5 inch double amplitude from 5 to 12.5 cps and a frequency that varies logarithmetically from 12.5 to 500 cps with accelerations maintained at, but not exceeding, 4 g's, At the two (2) most severe resonances encountered below 200 cycles per second, the mount shall be vibrated an additional 15 minutes ± 1 minute at 4 g's acceleration.

3.5.4 Shock. - The mount shall show no evidence of physical failure or damage and shall meet the requirements of 3.6 to 3.10 inclusive after being subjected to a total of 18 shocks of 100 g's peak amplitude with each shock pulse a terminal peak saw-tooth having a time duration of 6 ± 1 milliseconds. Three (3) shocks shall be applied in each direction along the three (3) mutually perpendicular axes of the mount. This requirement shall be met with the mount mounted as specified in 3.4(a and with the dummy load assembled as specified in 3.4(d).

3.6 Performance.

3.6.1 Boresight adjustment range.- With the mount oriented as specified in 3.4(a) and 3.4(b) and with the dummy load assembled as specified in 3.4(d), the elevation and azimuth screw assemblies shall allow each knob to be rotated a minimum of 2.5 revolutions between stops.

3.6.2 Horizontal looseness.- The total horizontal looseness shall not exceed .006 inch when measured at a point extending beyond the mount 12.5 inches from the axis of the tapered (king) pin (or 26.375 from the axis passing thru Yoke Pin 8616059 and Stud Pin 8616078). This requirement shall be met when a 20 lb. pressure is applied against the dummy load at the aforementioned distance and gradually released first in one direction then similarly applied and released in the opposite direction in a plane parallel to the mounting surface when the mount is mounted and oriented as specified in 3.4(a), 3.4(b), 3.4(c) and with the dummy load assembled as specified in 3.4(d).

3.6.3 Retention of elevation knob and dial settings.- The elevation knob and dial shall not change position by more than .0046 inch in a linear direction after the mount is subjected to the conditions specified in 3.5.3 and 3.5.4 (see 6.2). The elevation knob shall be positioned midway between limits of travel prior to subjecting the mount to the conditions specified in 3.5.3 and 3.5.4.

3.7 Operability.

3.7.1 Elevation and azimuth knobs.- The running torque required to rotate the elevation and azimuth knobs shall be 3 to 4 pound-inches (lb. in.) at any point within the limits of travel when the mount is mounted and positioned as specified in 3.4(a) and 3.4(b) and with the dummy load assembled as specified in 3.4(d). Similarly, the running



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torque shall not exceed 6 lb. in. at the-temperatures specified in 3.5.2. (NOTE: The weight of the dummy load may be relieved when meeting the elevation knob torque requirement.)

3.7.2 Elevation and azimuth dials.- The running torque required to rotate the elevation and azimuth dials shall be 16 to 20 ounce-inches (oz. in.) and shall not exceed 28 oz. in. at the temperatures specified in 3.5.2.

3.8 Electrical.

3.8.1 Insulation resistance.- With a potentioal of 100 volts direct current applied, insulation resistance shall not be less than 10 megohms.

3.8.2 Continuity. - The continuity of the wiring shall meet the applicable wiring and continuity requirements of Drawing c8616025.

3.9 Identification marking.- Identification marking shall be in accordance with MIL-STD-130 unless otherwise specified.

3.10 Workmanship. - In addition to meeting the applicable work-manship requirements of MIL-F-13926, the mounting surfaces and keys shall be free of paint, burrs or sharp edges that would preclude normal assembly or operation of the mount. Stud 8616078 and associ-ated tru-arc retaining ring shall be free of paint.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed require-ments.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The require-ment for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) mounts plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in



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regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is respnsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105$_0$

4.3.2 Examination and test.

4.3.2.1 Component and subassemblies.- Component and subassemblies shall be inspected in accordance with the inspection provisions contained in the applicable supplementary quality assurance provisions listed in the data list. All other components and subassemblies to be inspected in accordance with the Quality Assurance Provisions of MIL-F-13926.

4.3.2.2 First article test.- The mount shall successfully comply with the requirements outlined in Tables I and II. Additional first article testing will be required when the processes, material or changes in design occur subsequent to the original first article approval. All first articles so produced shall be submitted to the Government in accordance with 4.2 for approval.

4.3.2.3 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I and II. Examination and tests for Packaging, Packing, and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I and II shall constitute the minimum inspection to be performed by the Government after first article approval.



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TABLE I - CLASSIFICATION OF DEFECTS

| CLASS | REQUIREMENT | TEST PROCEDURE |
|---|---|---|
| CRITICAL: NONE | | |
| MAJOR: AQL 1.0% Defective | | |
| 101. Vibration "A" | 3.5.3.1 | 4.5.1.2 |
| 102. Horizontal Looseness | 3.6.2 | 4.5.2.2 |
| 103. Boresight Adjustment Range | 3.6.1 | 4.5.2.3 |
| 104. Torque elevation & azimuth knobs (60° to 90°F) | 3.7.1 | 4.5.2.4 |
| 105. Torque sliding dials (60° to 90°F) | 3.7.2 | 4.5.2.5 |
| 106. Insulation resistance | 3.8.1 | 4.5.2.6 |
| 107. Continuity | 3.8.2 | 4.5.2.7 |
| 108. Identification | 3.9 | 4.5.2.8 |
| 109. Workmanship | 3.10 | 4.5.2.9 |
| 110. Retention of Settings | 3.6.3 | 4.5.1.2, 4.5.1.3 & 4.5.1.4 |
| MINOR: | | |
| None | | |

4.3.2.4 Acceptance and rejection.- Rejection lots shall be screened for all defective characteristics. Removal correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.3 Certification.- Certification shall include test data and results for all characteristics specified below. Certification will be required prior to performing final acceptance inspection and shall suffice for the remainder of the production quantities, provided the manufacturing processes and techniques used to produce the items for which certification was issued have not been changed or revised. Any and all changes will require a new certification from the contractor.

| CHARACTERISTIC | TEST DATA TO COMPLY WITH |
|---|---|
| Fabrication | 3.1 |
| General specifications | 3.3 |

4.3,4 Special sampling.- Samples shall be selected at random from lots in accordance with inspection Level S-3 Table I of MIL-STD-105. Acceptance or rejection of the lot shall be in accordance with AQL 2.5 of Table II-A of MIL-STD-105. Samples shall have met all the requirements and tests outlined in Table I prior to being subjected to the following examination and tests.

TABLE II

| CHARACTERISTIC | REQUIREMENT | TEST PROCEDURE |
|---|---|---|
| 301. Shock | 3.5.4 | 4.5.1.4 |
| 302. Torque elevation and azimuth knobs (-40° + 150°F) | 3.7.1 | 4.5.2.4 |
| 303. Torque elevation and azimuth dials (-40° + 150°F) | 3.7.2 | 4.5.2.5 |
| 304. Temperature (Storage -80°+160°F) | 3.5.1 | 4.5.1.1 |
| 305. Temperature (Operating -40° + 150°F) | 3.5.2 | 4.5.1.1 |
| 306. Vibration "B" | 3.5.3.2 | 4.5.1.3 |



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4.3.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.2.4).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.4.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.4,2 Contractor furnished inspection equipment.

4.4.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications forming a part of the contract shall be supplied by the contractor.

4.4.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with “Test Methods and Procedures” specified in 4.5 of this specification and with the component inspection procedures specified in “Examination” and "Test Facilities” requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimesnions and tolerances used in the design of test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Temperature.- The mount shall be subjected to Test Procedure 11 of MIL-F-13926. However, the temperatures specified in 3.5.1 and 3.5.2 shall apply. When the mount has been stabilized at the high and low temperatures specified in 3.5.1 and 3.5.2 the torque of knobs and scales shall be inspected for conformance with 3.7.1 and 3.7.2. After having been stabilized at standard ambient temperatures, the mount shall-be inspected for conformance with 3.2.

4.5.1.2 Vibration. - The mount shall be vibrated in accordance with Test Procedure V of MIL-F-13926. However, the frequency, amplitude and duration specified in 3.5.3.1 shall apply. After



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completion of the vibration test, the mount shall be inspected for conformance with 3.5.3.

4.5.1.3 Vibration "B".- This test shall be performed as specified in MIL-STD-810, Method 514, Procedure I, with the following exceptions: the amplitued, frequency, acceleration and duration shall be as specified in 3.5.3.2. At the conclusion of this test, the mount shall be inspected by visual and tactile means for damage, then successfully tested for compliance with the performance requirements specified in 3.5.3.

4.5.1.4 Shock. - This test shall be performed as specified in MIL-STD-810, Method 516, Procedure IV, with the amplitude, duration, number of shocks and direction and with cylindrical gage F10553L52 assembled to the mount to represent the requirement specified in 3.5.4 At the conclusion of this test, the mount shall be inspected by visual and tactile means for evidence of damage, then successfully tested for compliance with the performance requirements specified in 3.5.4.

4.5.2 Performance.

4.5.2.1 Orientation. - The mount will be considered properly orientated when position in a testing medium having the accuracy depicted on drawing F11739343 and when properly secured by means of 8 bolts supplied with the mount. Insert a testing medium having the accuracy depicted on drawing F10553152 into the receiving diameter of the mount and by means of the taper pin supplied with the mount,secure the testing medium to the mount. Set the azimuth and elevation knobs at the mid point of travel. Set the indicators as outlined in the set-up instructions.

4.5.2.2 Horizontal looseness.- With the mount positioned as outlined in 4.5.2.1 apply a 20 lb. force in the direction specified in 3.6.2. Total horizontal looseness as measured by the indicator shall not exceed the limits specified in 3.6.2. Total looseness is measured after applying a 20 lb. load on the test medium secured in the mount in a direction parallel to the mounting surface as specified in 3.6.2.

4.5.2.3 Boresight adjustment range.- With the mount positioned as outlined in 4.5.2.1, use visual and tactile inspection techniques, rotate the azimuth adjustment knob between both stops in both clockwtse and then counterclockwise direction. The number of rotations in each direction shall be as outlined in 3.6.1. Repeat this test on the elevation adjustment knob.

4.5.2.4 Torque elevation and azimuth knob.- With the mount positioned as outlined in 4.5.2.1, torque shall be measured from the mid point of travel in the clockwise and counterclockwise direction of azimuth and elevation adjustment knobs. The test to be performed using appropriate adapter and standard shop type torque wrench. All running torque measurements to be within the limits outlined in 3.7.1. This test also to be performed at the operating temperature



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outlined in 3.5.2.

4.5.2.5 Torque azimuth and elevation dials.- With the mount Positioned as outlined in 4.5.2.1 torque shall be measured using standard shop type torque adapter and Wrench. All running torque measurements shall be within the limits of 3.7.2 at each temperature specified.

4.5.2.6 Insulation resistance. - This test to be performed as outlined in Test Method 302 of MIL-STD-202 using Test Condition A. Test results shall comply with 3.8.1.

4.5.2.7 Continuity.- Using standard shop type continuity checker determine the requirement specified in 3.8.2 has been met.

4.5.2.8 Identification markings.- Use test methods of MIL-STD-130 to determine that marking conforms with the requirements of 3.9.

4.5.2.9 Workmanship.- Workmanship shall be in accordance with MIL-F-13926 and by means of visual and tactile inspection techniques determine that the requirement for workmanship has been met.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing and marking shall be in accordance with Packaging Data Sheet No. 8616011. The level of protection shall be as specified in the procurement document.

## 6. NOTES

6.1 Intended use. - The Telescope Mount, M146 is mounted on the right side of the 155mm Self-Propelled Howitzer M109 and the 105mm Self-Propelled Howitzer M108. It is used to support a direct fire telescope and provides azimuth and elevation boresight corrections.

6.2 Retention of elevation knob and dial.- The .0046 inch linear movement of the knob is equivalent to an angular movement of 0.3 mil measured from the pivot point at the 3.5625 diameter.

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Preparing activity:

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Project No. 1240-A387

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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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